# AI Operations Audit: Top 3 Time Drains

Mid-sized third-party logistics provider · Kanto region, Japan · ~60 office staff · data period Nov 2025 – Oct 2026

## Executive summary

The office team logged **104,555 hours** in 12 months, a loaded staff cost of **¥355.9M**. Management asked which three operational time drains to fix first with AI and what that is worth. We measured time from the task-tracking system, cross-checked it against ticket, document and invoice data, and compared it with what staff believe.

| 25% | 9.9 FTE | ¥51.0M | 3.3 mo |
|---|---|---|---|
| of all hours go to the top 3 drains | capacity freed (17,287 h/yr) | net annual benefit after running costs | blended payback on ¥14.0M one-off |

### The three drains to fix first

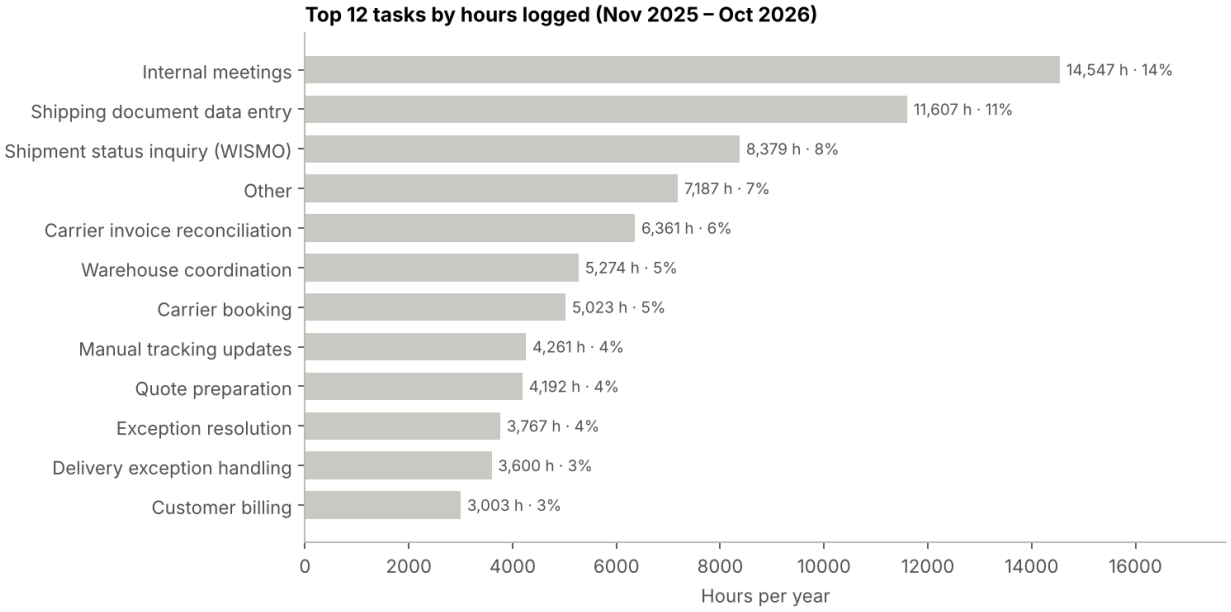
| # | Time drain | Hours / yr | Share | Recommended AI lever |
|---|---|---|---|---|
| 1 | Shipping document data entry | 11,607 | 11% | Intelligent document processing (OCR + LLM extraction), human validation |
| 2 | Shipment status inquiry (WISMO) | 8,379 | 8% | Proactive tracking notifications + LLM reply assistant on live TMS data |
| 3 | Carrier invoice reconciliation | 6,361 | 6% | Automated matching against contract rate cards; people handle exceptions |

### Key messages

- **Meetings are the biggest block of time (14%) but not the biggest AI opportunity.** AI can only trim them. The top 3 drains together take 25% of hours, nearly twice as much.
- **Staff perception points the wrong way.** People over-weight meetings and Excel reporting, and under-weight document keying, status inquiries and invoice checking, which are repetitive work spread thinly through the day.
- **Each drain has a measurable symptom:** 24% of keyed documents contain an error; 46% of labelled tickets ask "where is my shipment?" and 22% of those are reopened; ¥7.8M of carrier overbilling (38%) was never recovered.
- **The business case holds under stress.** At half the planned automation, the programme still returns ¥21.3M a year and pays back in 8 months.
- **Start with WISMO** (fastest to launch, visible to customers), then invoice matching, then document processing.

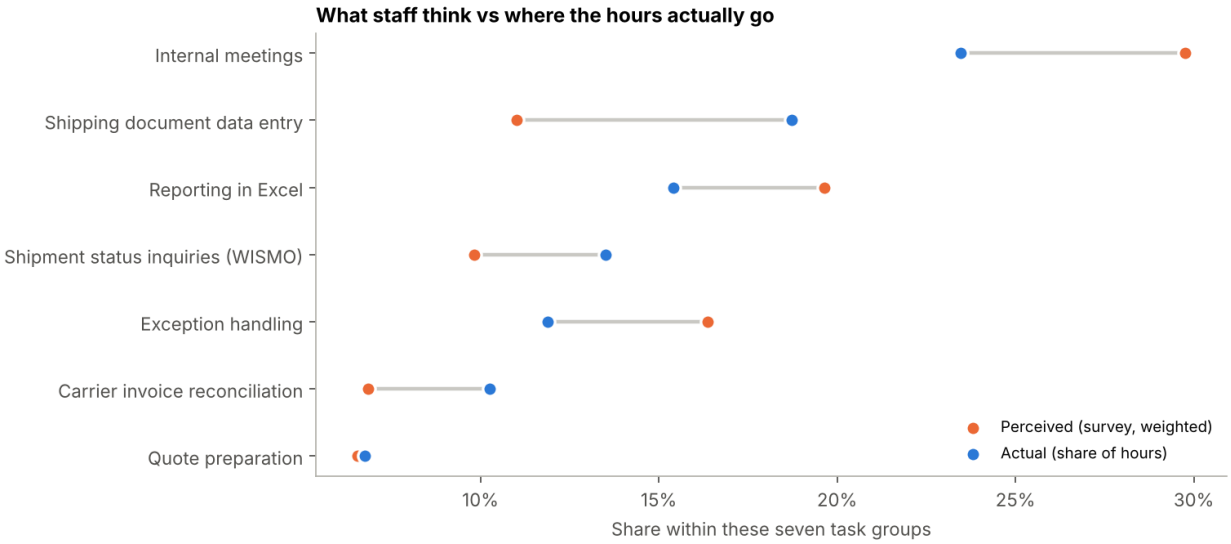
## 1. Where the time goes

Time-tracking entries were cleaned first (duplicate submissions, 100+ spellings of 25 task types, missing and runaway timers, mixed date formats) so that a typical day sums to 7–8.5 hours. Hours were then costed with each employee's loaded hourly rate.



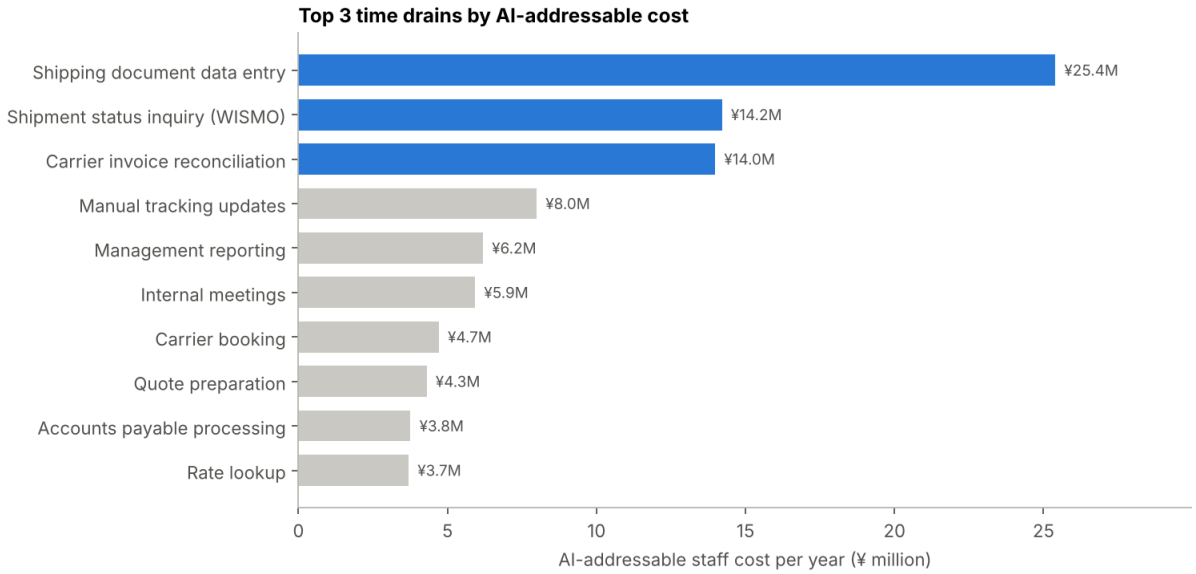

## 2. Perception vs data

Before seeing the data, 56 staff named their three biggest time drains (weighted 3-2-1). Visible, frustrating work is over-weighted; high-volume micro-tasks are under-weighted. Shipping document entry shows the largest gap.



## 3. Which drains AI can actually remove

For every task we estimated the share of time AI or automation can realistically remove within 12 months, based on how repetitive, rule-based and digital the work is. Multiplying by the cost of the hours gives the **AI-addressable cost**, which is the ranking used to pick the top 3. The percentages are analyst assumptions and are listed so they can be challenged.




| Task | Hours / yr | AI share | Addressable cost / yr | Lever |
|---|---|---|---|---|
| Shipping document data entry | 11,607 | 70% | ¥25.4M | Intelligent document processing (OCR + LLM extraction) with human validation |
| Shipment status inquiry (WISMO) | 8,379 | 60% | ¥14.2M | Proactive tracking notifications + LLM auto-reply grounded in TMS data |
| Carrier invoice reconciliation | 6,361 | 65% | ¥14.0M | Automated rate-card matching with anomaly flags; humans handle exceptions only |
| Manual tracking updates | 4,261 | 60% | ¥8.0M | Carrier API / EDI integration (automation, not AI) |
| Management reporting | 2,411 | 40% | ¥6.2M | Automated BI dashboards replace copy-paste Excel reports |
| Internal meetings | 14,547 | 10% | ¥5.9M | AI meeting notes and action items (limited impact) |

Manual tracking updates (#4) share a root cause with WISMO, the lack of an automated tracking feed, so they are addressed by the same initiative.

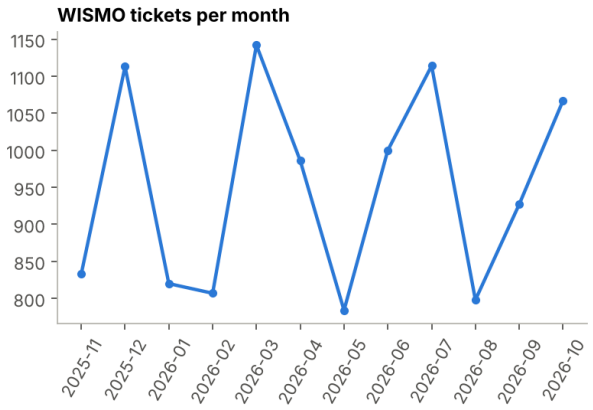
# 4. Deep dives and recommendations

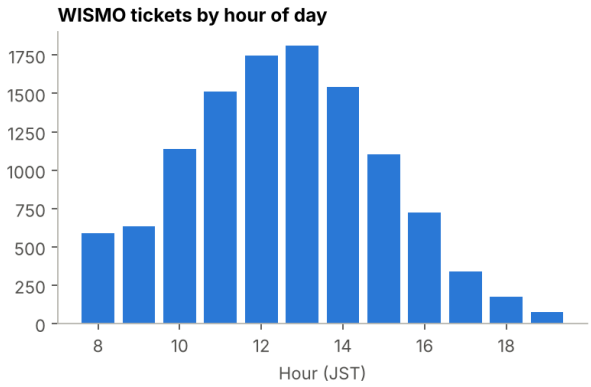
## 4.1 Shipping document data entry (Operations)

**Finding.** 41,000 documents a year are re-typed into the WMS. 24% contain at least one keying error, creating 2,042 hours of rework; error rates rise with document complexity (customs declarations 39%). 75% arrive as PDF or scans. **Recommendation.** Intelligent document processing extracts fields, a person validates low-confidence fields, and the result is pushed to the WMS. Excel files are imported directly. Target: ~70% of keying time removed.

## 4.2 Shipment status inquiries / WISMO (Customer Service)

**Finding.** 46% of labelled tickets are status questions. Each takes ~10 minutes, but volume peaks in December and March, and 22% are reopened because the customer chases again (median first response 94 min). The answer already exists in the TMS. **Recommendation.** (1) Proactive tracking emails/SMS at key milestones; (2) an LLM reply assistant that answers WISMO emails from live tracking data, introduced in suggest-mode first.



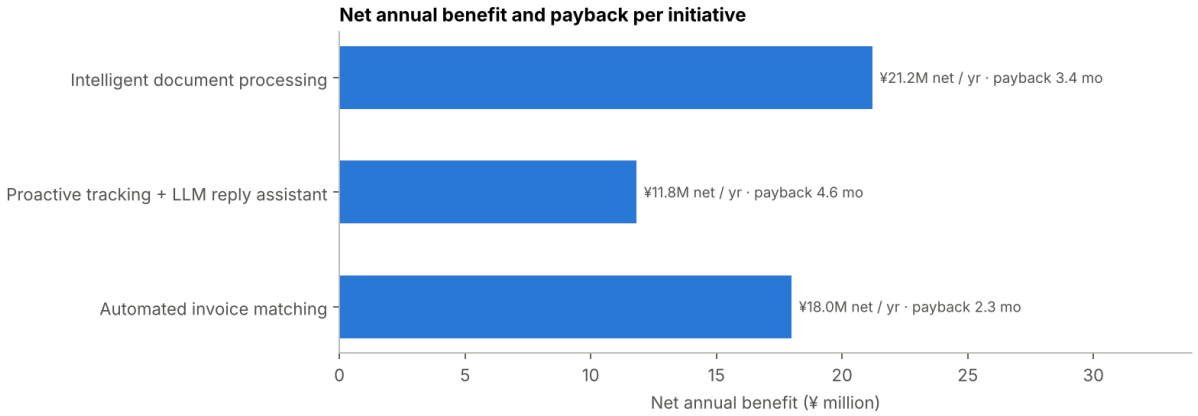

## 4.3 Carrier invoice reconciliation (Finance)

**Finding.** All 34,000 carrier invoice lines are checked by hand, yet only ~14% are wrong. Overbilling totalled ¥20.5M, mostly fuel surcharges and wrong weight tiers, and 38% of it was never recovered. **Recommendation.** Match every line automatically against the contract rate card, flag mismatches by reason, and let people work only the exceptions. Earlier, better-evidenced disputes raise recovery (assumed 62% to 90%).

## 5. Business case

Value = hours freed × average loaded hourly cost of the staff doing the task, plus improved overbilling recovery for invoices. Costs are typical estimates for a company of this size (software plus integration) and should be replaced with vendor quotes.

| Initiative | Hours freed | FTE | Gross benefit | One-off | Running / yr | Net / yr | Payback |
|---|---|---|---|---|---|---|---|
| Intelligent document processing | 8,125 | 4.7 | ¥25.4M | ¥6.0M | ¥4.2M | ¥21.2M | 3.4 mo |
| Proactive tracking + LLM reply assistant | 5,027 | 2.9 | ¥14.2M | ¥4.5M | ¥2.4M | ¥11.8M | 4.6 mo |
| Automated invoice matching | 4,135 | 2.4 | ¥19.8M | ¥3.5M | ¥1.8M | ¥18.0M | 2.3 mo |
| **Total** | **17,287** | **9.9** | **¥59.4M** | **¥14.0M** | **¥8.4M** | **¥51.0M** | **3.3 mo** |




### Sensitivity

| Scenario | Net annual benefit | Payback |
|---|---|---|
| 50% of planned automation | ¥21.3M | 7.9 months |
| 75% of planned automation | ¥36.1M | 4.6 months |
| 100% of planned automation | ¥51.0M | 3.3 months |

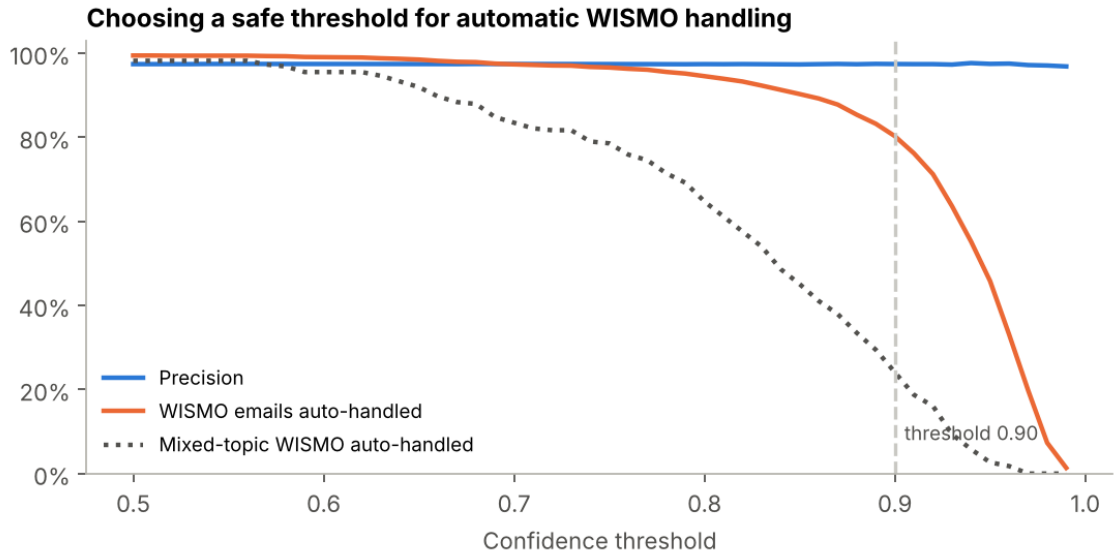
Freed hours are capacity, not automatic cash savings. The value is realised by absorbing volume growth without new hires, reducing overtime in peak months and moving staff to higher-value work such as exception handling and customer care.

## 6. Roadmap

| Phase | Timing | What | Success measure |
|---|---|---|---|
| 1 · Quick win | Month 0–3 | Proactive tracking notifications; WISMO reply assistant in suggest-mode; auto-send for high-confidence emails after a 4-week pilot | WISMO tickets −40%, reopen rate below 10% |
| 2 · Finance | Month 2–5 | Rate-card matching for the two largest carriers, then all five; dispute workflow | Lines checked by hand −65%; recovery rate ≥ 85% |
| 3 · Documents | Month 4–9 | Document processing for delivery notes and packing lists first, then invoices, B/L and customs; WMS integration | Keying time −70%; error rate below 5% |
| Enablers | Ongoing | Owner per initiative, monthly time-tracking review, staff training, data-privacy check for customer emails (APPI) | Adoption above 80% of target users |

## 7. Proof of concept: email classifier

To test the WISMO lever, we trained a TF-IDF + logistic regression model on the historic inbox. On a hold-out set it reached 97.4% accuracy (macro-F1 0.97) versus 0.87 macro-F1 for keyword rules. With a conservative confidence threshold of 0.90 it can auto-handle about 80% of WISMO emails at ~97% precision, while most mixed-topic emails stay with a person. Precision is capped by the ~3% of historic agent labels that are wrong.




## 8. Assumptions and limitations

- Time tracking reflects how staff record their work; under-recording of short interruptions is likely, so high-volume tasks may be understated rather than overstated.
- AI-addressable percentages and costs are estimates; the sensitivity table shows the impact of missing them.
- The classifier was tested on historic emails only; a live pilot is required before auto-sending replies.
- This report is a portfolio case study built on realistic synthetic data; no real company data is used.